Topology of Neural-Network Layers for Out-of-Distribution Detection: A Unified View and Benchmark of Six Topological Approaches

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Abstract

Deep classifiers are often highly confident on inputs that come from outside their training distribution. Topological data analysis (TDA) offers an alternative signal: instead of the network's output confidence, it summarizes the *shape* of the network's internal activity with persistent homology. Several such approaches have been proposed, but they use different objects (weight graphs, activation graphs, Bayesian samples, point clouds of activations), different summaries, and different evaluation protocols. We place six of them in one framework — *object* → *filtration* → *persistence diagram* → *score* — implement all of them on shared models, and evaluate them as out-of-distribution (OOD) detectors under a common protocol. Our study has two parts: (i) a controlled study with a Bayesian MLP trained on four in-distribution datasets (MNIST, Fashion-MNIST, KMNIST, SVHN), each tested against 10–12 OOD sets grouped into near-OOD, far-OOD and covariate shift; and (ii) the OpenOOD v1.5 CIFAR-10 and CIFAR-100 benchmarks with the released ResNet-18 checkpoints, where our pipeline reproduces the published baselines exactly. We find that (1) on the MLPs, topological scores reliably detect the inputs on which softmax confidence fails, but on the well-trained ResNet-18 no per-image topological score beats the output-based baselines; (2) no per-image topological score wins on every dataset, while point-cloud topology on small batches of inputs is strong whenever the classes are few; (3) simple summaries (β_0 curves, 0-dimensional features) often beat richer ones (full persistent-Laplacian spectra, H_1 features); and (4) how layers are combined, and which layers are available, matter as much as the summary itself.

1 Introduction

A classifier trained on a fixed set of classes will still produce a prediction for an image from an unseen class, and often a confident one. Detecting such *out-of-distribution* (OOD) inputs is a prerequisite for deploying neural networks in safety-critical settings, and has become a large research area with standardized benchmarks such as OpenOOD [20, 22].

Most post-hoc OOD detectors read a scalar from the network's output or penultimate features: the maximum softmax probability [6], an energy [10], a feature-space distance [16]. A different line of work asks what the *topology* of the network's internal activity reveals. Rieck et al. [15] summarize trained weights by the persistent homology of layer graphs; Lacombe et al. [9] build one activation graph per input and layer and compare its persistence diagram with class-wise references; Yeh and Yang [21] extend this to Bayesian networks; persistent Laplacians [2, 8, 19] add spectral, geometric information to persistent homology; and Magai and Ayzenberg [11] and Purvine et al. [14] study the topology of point clouds of hidden activations.

These approaches were proposed with different goals (model selection, uncertainty, interpretability), objects and evaluation protocols, so it is hard to tell which ideas are useful for OOD detection. This draft makes three contributions:

- **A unified framework** (Section 3) that writes all six approaches as *object* → *filtration* → *persistence diagram* → *score*, with the equations and a diagram for each (Section 4).
- **A controlled multi-dataset study** (Section 6): every approach on the same Bayesian MLP trained on four in-distribution (ID) datasets, three seeds each, against near-OOD, far-OOD and covariate-shift sets.
- **A benchmark on OpenOOD v1.5** (Section 7): CIFAR-10 and CIFAR-100 with the released ResNet-18 checkpoints, reported next to the published post-hoc methods.

2 Background: persistent homology and persistent Laplacians

Filtrations and persistence diagrams. A *filtration* is a nested family of simplicial complexes $K_t \subseteq K_{t'}$ for $t \le t'$. Persistent homology tracks the q-dimensional homology classes ($q = 0$: connected components, $q = 1$: loops) through the filtration. A class born at b and dying at d gives a point (b, d) of the *persistence diagram* $\mathcal{D}_q$, a multiset in the extended plane; its *persistence* is $|d - b|$. For a weighted graph $G = (V, E, w)$, the *superlevel filtration* adds all vertices first and then edges in decreasing order of weight,

$$G^{(t)} = (V,\ \{e \in E : w(e) \ge t\}), \qquad t \text{ decreasing.} \tag{1}$$

Since every vertex exists from the start, only 0-dimensional features change, and each edge that merges two components contributes one point. These edges are exactly those of a *maximum spanning tree* (MST) of G, so

$$\mathcal{D}_0(G) = \{w(e) : e \in \mathrm{MST}(G)\}, \tag{2}$$

a multiset of $|V| - 1$ real numbers (the birth of every class is the top of the filtration). For a point cloud $X = \{x_1, \dots, x_n\} \subset \mathbb{R}^d$, the *Vietoris–Rips* filtration

$$\mathrm{VR}_\varepsilon(X) = \{\sigma \subseteq X : \|x_i - x_j\| \le \varepsilon \ \ \forall x_i, x_j \in \sigma\} \tag{3}$$

yields diagrams in every dimension; H_0 deaths are again MST edge lengths, now of the Euclidean graph on X.

Distances between diagrams. The p-Wasserstein distance between diagrams matches points optimally, allowing points to be matched to the diagonal Δ:

$$W_p(\mathcal{D}, \mathcal{D}') = \Big(\inf_{\gamma : \mathcal{D} \cup \Delta \to \mathcal{D}' \cup \Delta} \sum_u \|u - \gamma(u)\|_\infty^p \Big)^{1/p}. \tag{4}$$

For the 1-D diagrams of Eq. (2) with a fixed number of points, the optimal matching sorts both multisets, so W_2 is the Euclidean distance between sorted vectors. The *sliced Wasserstein* distance [3] averages 1-D Wasserstein distances of projections onto lines ℓ_θ:

$$\mathrm{SW}(\mathcal{D}, \mathcal{D}') = \frac{1}{\pi} \int_{-\pi/2}^{\pi/2} W_1\big(\pi_\theta(\mathcal{D} \cup \pi_\Delta(\mathcal{D}')),\ \pi_\theta(\mathcal{D}' \cup \pi_\Delta(\mathcal{D}))\big)\, d\theta, \tag{5}$$

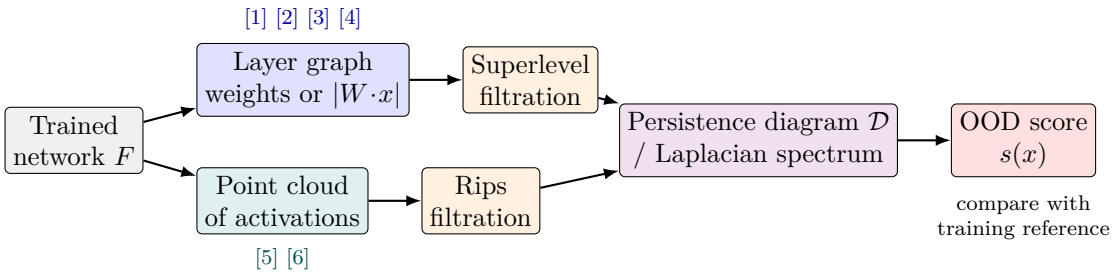


Figure 1: All six approaches follow one pipeline: build an object from the network, filter it, summarize it with a persistence diagram (or a persistent-Laplacian spectrum), and turn the summary into an OOD score by comparing it with what training data produce.

	Object	Depends on x?	Summary	OOD score
[1] NP [15]	graph of $\lvert W_\ell \rvert$	no	$\lVert \mathcal{D}_0 \rVert_p$	(input-dependent variant, §4.1)
[2] TU [9]	graph of $\lvert W_\ell\, x_{\ell-1} \rvert$	yes	$\mathcal{D}_0$ (MST)	W_2 to class-mean $\mathcal{D}_0$
[3] pTU [21]	same, S posterior samples	yes	$\mathcal{D}_0$	Fréchet variance; permutation test
[4] PL [19]	thresholded activation graph	yes	spectrum of $\Delta_0^{K,L}$	distance to class-mean spectrum
[5] Manifold [11]	point cloud per layer	set of inputs	H_0, H_1, intrinsic dim.	batch descriptors; local ΔH_0
[6] CNN act. [14]	point cloud per layer	set of inputs	$\mathcal{D}_0, \mathcal{D}_1$	SW to reference; layer profile

Table 1: The six approaches in the framework of Fig. 1.

where π_θ projects onto ℓ_θ and π_Δ onto the diagonal. The *Fréchet mean* and *variance* of diagrams $\mathcal{D}^{(1)}, \dots, \mathcal{D}^{(S)}$ are

$$\bar{\mathcal{D}} = \arg\min_{\mathcal{D}} \sum_s \mathrm{Dist}(\mathcal{D}^{(s)}, \mathcal{D})^2, \qquad \mathrm{Var} = \frac{1}{S} \sum_s \mathrm{Dist}(\mathcal{D}^{(s)}, \bar{\mathcal{D}})^2. \tag{6}$$

For equal-size 1-D diagrams under W_2, $\bar{\mathcal{D}}$ is the coordinate-wise mean of the sorted vectors.

Persistent Laplacians. Let ∂_q^K be the boundary map of a complex K and $\Delta_q^K = \partial_{q+1}^K \partial_{q+1}^{K\top} + \partial_q^{K\top} \partial_q^K$ its combinatorial Laplacian. For a pair $K \subseteq L$, let $C_{q+1}^{L,K} = \{c \in C_{q+1}^L : \partial_{q+1}^L c \in C_q^K\}$ and let $\partial_{q+1}^{L,K}$ be the restriction of ∂_{q+1}^L to it. The *persistent Laplacian* is [12, 18]

$$\Delta_q^{K,L} = \underbrace{\partial_{q+1}^{L,K} (\partial_{q+1}^{L,K})^\top}_{\Delta_{q,\mathrm{up}}^{K,L}} + \underbrace{(\partial_q^K)^\top \partial_q^K}_{\Delta_{q,\mathrm{down}}^{K}}. \tag{7}$$

Its kernel has dimension equal to the persistent Betti number $\beta_q^{K,L}$, so it contains the information of persistent homology; its nonzero eigenvalues add geometric information [12]. Wolf et al. [19] prove that the up and down parts are stable while the full operator need not be.

3 A unified framework

Figure 1 shows the common structure. Let $F : \mathbb{R}^{h_0} \to \mathbb{R}^K$ be an L-layer network, x_ℓ the input of layer ℓ (so $x_0 = x$), and $W_\ell \in \mathbb{R}^{h_\ell \times h_{\ell-1}}$ its weights. The approaches differ in (a) the object: a bipartite layer graph or a point cloud; (b) whether the object depends on the input x; and (c) how the diagram becomes a score. Table 1 summarizes them; Section 4 gives the details.

4 The six approaches

4.1 [1] Neural persistence

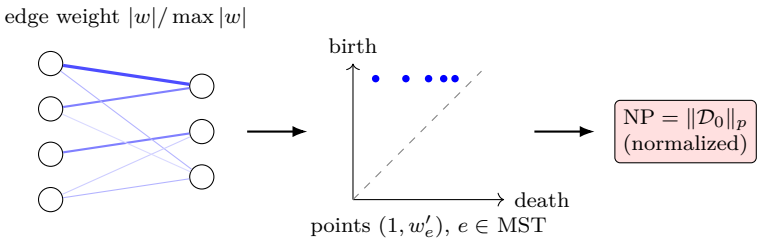


Figure 2: [1] Neural persistence: a layer of trained weights as a bipartite graph, its 0-dimensional diagram (the MST weights), and the p-norm of the diagram.

Rieck et al. [15] view layer ℓ as the complete bipartite graph $G_\ell = (V_{\ell-1} \sqcup V_\ell, E_\ell)$ with weights normalized over the whole network,

$$w'(i,j) = \frac{|W_\ell(j,i)|}{\max_{\ell',i',j'} |W_{\ell'}(j',i')|} \in [0,1]. \tag{8}$$

Under the superlevel filtration (Eq. (1)) every vertex is born at 1 and every MST edge e kills a component at w'_e, giving $\mathcal{D}_0(G_\ell) = \{(1, w'_e)\}$. Neural persistence is

$$\mathrm{NP}(G_\ell) = \|\mathcal{D}_0(G_\ell)\|_p = \Big(\sum_{e \in \mathrm{MST}(G_\ell)} (1 - w'_e)^p \Big)^{1/p}, \qquad 0 \le \mathrm{NP}(G_\ell) \le (1 - \min_e w'_e)\,(|V_\ell| - 1)^{1/p}, \tag{9}$$

normalized by the upper bound and averaged over layers. NP depends only on the weights, so it assigns the same value to every input and cannot rank inputs (AUROC $= 50$ by construction). *Adaptation.* We apply Eq. (9) to the input-dependent activation graph of [2] (Eq. (10)) and score $s(x) = |\,\mathrm{NP}(x) - \mu_k|/\sigma_k$, where (μ_k, σ_k) are the training statistics of the predicted class $k = \arg\max F(x)$.

4.2 [2] Topological uncertainty

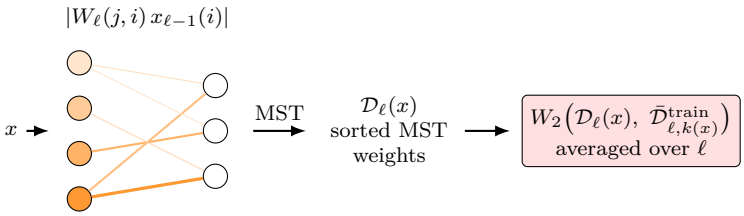


Figure 3: [2] Topological uncertainty: one activation graph per input and layer; its diagram is compared with the Fréchet mean of training diagrams of the predicted class.

Lacombe et al. [9] weight the edges of layer ℓ by how much input x uses them,

$$w_x(i,j) = |W_\ell(j,i)\, x_{\ell-1}(i)|, \tag{10}$$

which gives one *activation graph* $G_\ell(x)$ per input and layer. Its diagram $\mathcal{D}_\ell(x)$ is the multiset of its $N_\ell = h_{\ell-1} + h_\ell - 1$ MST weights (Eq. (2)). For each class k, the reference $\bar{\mathcal{D}}^{\text{train}}_{\ell,k}$ is the Fréchet mean (Eq. (6)) of the training diagrams predicted as k, and

$$\text{TU}(x) = \frac{1}{L}\sum_{\ell=1}^{L} W_2\big(\mathcal{D}_\ell(x),\ \bar{\mathcal{D}}^{\text{train}}_{\ell,k(x)}\big), \qquad k(x) = \arg\max_k F(x)_k. \tag{11}$$

Variant (ours). The layer distances live on different scales, so the plain average is dominated by the largest. *TU-rescaled* divides each layer's distance by its mean over training inputs before averaging.

4.3 [3] Predictive topological uncertainty

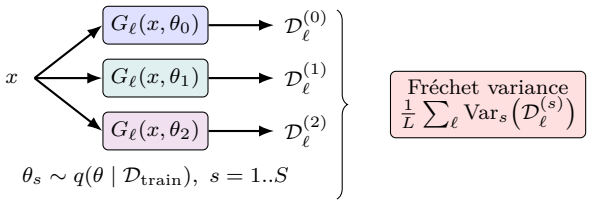


Figure 4: [3] pTU: the activation-graph diagrams of S posterior samples of a Bayesian network; their spread is the uncertainty.

For a Bayesian network with (approximate) posterior $q(\theta \mid \mathcal{D}_{\text{train}})$, Yeh and Yang [21] draw $\theta_1, \ldots, \theta_S$ and build $G_\ell(x, \theta_s)$ as in Eq. (10), each with diagram $\mathcal{D}_\ell^{(s)}(x)$. The predictive topological uncertainty is the posterior Fréchet variance averaged over layers,

$$\text{pTU}(x) = \frac{1}{L}\sum_{\ell=1}^{L}\frac{1}{S}\sum_{s=1}^{S} W_2\big(\mathcal{D}_\ell^{(s)}(x),\ \bar{\mathcal{D}}_\ell(x)\big)^2. \tag{12}$$

At the dataset level, the paper tests $H_0 : P_{\text{test}} = P_{\text{train}}$ with a permutation test whose statistic is the empirical Wasserstein distance between the pTU values of m training and m test inputs, $T = \big(\sum_{i=1}^{m} |Y_{(i)} - Z_{(i)}|^p\big)^{1/p}$, with order statistics $Y_{(i)}, Z_{(i)}$. *Posterior.* For the MLP we train with Bayes-by-Backprop [1]; for the ResNet-18 checkpoints we use a diagonal last-layer Laplace approximation [4], $q(W) = \mathcal{N}(W^\star, (\tau I + \text{diag}\,\text{GGN})^{-1})$, with the prior precision τ chosen on the validation set. *Variant (ours). pTU two-sided* scores $|\log \text{pTU}(x) - \text{median}_{\text{train}} \log \text{pTU}|$, because we find that OOD inputs can have *lower* pTU than training data.

4.4 [4] Persistent Laplacian

Proposition 1. *Let K_t be the superlevel filtration (Eq. (1)) of a graph whose vertices are all present at the start. For $K_s \subseteq K_t$, $\Delta_0^{K_s,K_t} = L(K_t) = D_t - A_t$, the combinatorial graph Laplacian of K_t. For a bipartite graph, $\Delta_q^{K_s,K_t} = 0$ for $q \geq 1$.*

Sketch. $\Delta_{0,\text{down}} = 0$ since $\partial_0 = 0$. Every edge of K_t has both endpoints in $V = V(K_s)$, so $C_1^{K_t,K_s} = C_1^{K_t}$ and $\Delta_{0,\text{up}}^{K_s,K_t} = \partial_1^{K_t}\partial_1^{K_t\top} = L(K_t)$. A bipartite graph has no triangles, so there are no 2-simplices and no 1-cycles bound by them. □

Hence the zero eigenvalues of $L(K_t)$ count $\beta_0(K_t)$, which is the information of the diagram of [2], while the nonzero eigenvalues are what the persistent Laplacian adds. For three thresholds t

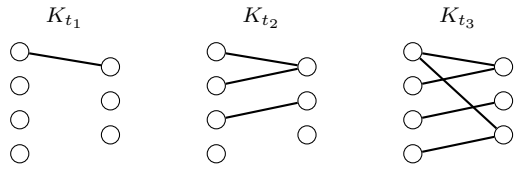


Figure 5: [4] Persistent Laplacian on the activation graph: in dimension 0 with all vertices born first, $\Delta_0^{K_s,K_t}$ is the graph Laplacian of K_t.

(quantiles of training MST weights) we use $\phi_t(x) = (\beta_0,\ |E(K_t)|,\ \lambda^+_{\min},\ \lambda_{\max},\ \mathrm{std}(\lambda^+))$ and score

$$s_{\mathrm{PL}}(x) = \frac{1}{|\mathcal{T}|\,|\phi|} \sum_{\ell,t} \left\| \frac{\phi_{\ell,t}(x) - \mu_{\ell,t,k(x)}}{\sigma_{\ell,t}} \right\|_2^2, \tag{13}$$

with class-wise means μ and standard deviations σ from training data. Using only the β_0 coordinate isolates the persistent-homology part.

4.5 [5] Topology and geometry of the data manifold

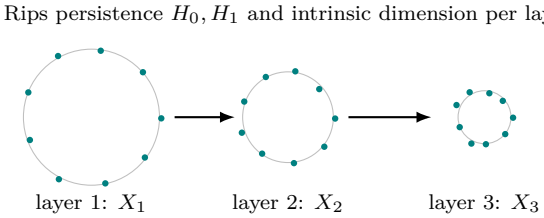


Figure 6: [5] Each layer maps a set of inputs to a point cloud; its topology and dimension are tracked across layers.

For inputs $x_1, \ldots, x_n$ and layer ℓ, Magai and Ayzenberg [11] study $X_\ell = \{h_\ell(x_1), \ldots, h_\ell(x_n)\} \subset \mathbb{R}^{h_\ell}$ (for convolutional layers, spatially averaged), its Rips diagrams $\mathcal{D}_0, \mathcal{D}_1$ (Eq. (3)), the *total persistence* $P_q(X_\ell) = \sum_{(b,d)\in\mathcal{D}_q}(d-b)$ and an intrinsic-dimension estimate; we use TwoNN [5],

$$\hat{m}(X) = \frac{n}{\sum_{i=1}^n \log(r_2(x_i)/r_1(x_i))}, \tag{14}$$

with r_1, r_2 the distances to the two nearest neighbours. *Batch score.* A batch of m inputs gives $\psi = (P_0, P_1, \hat{m})_{\ell=1..L}$, scored by its mean squared z-score against training batches of the same size. *Per-image adaptation (ours).* For an input x let $N_k(x)$ be its k nearest training points in layer ℓ; the local change

$$\Delta P_0^\ell(x) = P_0(N_k(x) \cup \{h_\ell(x)\}) - P_0(N_k(x)) \tag{15}$$

is the increase of the minimum-spanning-tree length when x joins its neighbourhood; the score is the mean squared z-score of $(\Delta P_0^\ell)_\ell$.

4.6 [6] Topology of CNN activations

Purvine et al. [14] represent each image at a convolutional layer by the channel vector at the spatial position of largest activation norm, compute the Rips diagrams of the resulting point cloud, and

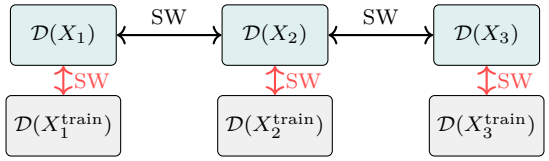


Figure 7: [6] Diagrams of every layer, compared across layers (Purvine et al.'s layer profile) and with reference diagrams of training batches (our OOD score).

compare layers with the sliced Wasserstein distance (Eq. (5)). We use two batch scores: the distance to reference diagrams, $s_{\text{SW}}(B) = \frac{1}{2L}\sum_{\ell,q} z\big(\frac{1}{R}\sum_r \text{SW}(\mathcal{D}_q(X_\ell^B), \mathcal{D}_q(X_\ell^r))\big)$ over R training batches, and the *layer profile* $\big(\text{SW}(\mathcal{D}_q(X_\ell^B), \mathcal{D}_q(X_{\ell+1}^B))\big)_{\ell,q}$ scored by its squared z-score.

5 Experimental protocol

Metrics. Following OpenOOD [22], OOD is the positive class. We report AUROC and FPR@95 (the fraction of ID inputs flagged at the threshold that detects 95% of OOD inputs); each is computed per OOD set and averaged within the near-OOD and far-OOD groups. All numbers are means (ś std) over three training seeds or checkpoints.

Per-image and batch settings. The standard setting scores one input at a time. Since [3], [5] and [6] are defined on sets of inputs, we also score batches of $m \in \{10, 20\}$ inputs drawn from a single source and report AUROC over batches.

Baselines. MSP [6], energy (EBO) [10], max logit (MLS) [7], KNN [16], ViM [17], and the Bayesian mutual information of the predictive distribution.

6 Part I: a controlled study across in-distribution datasets

Setup. We use the Bayesian MLP 784–64–16–10 of [21], trained with Bayes-by-Backprop (scale-mixture prior $\sigma_1 = 1, \sigma_2 = 0.007$, KL term scaled by 0.1, early stopping with patience 25), on four ID datasets, all grayscale 28×28 in $[0, 1]$. OOD sets come from OpenOOD's MNIST benchmark (NotMNIST, Fashion-MNIST, Textures, CIFAR-10, TinyImageNet, Places365) plus SVHN, KMNIST, Gaussian and uniform noise; covariate-shift sets are MNIST-C [13] and pixel-flip corruptions [9]. Table 2 lists the groups.

Findings.

- **Softmax confidence is the weakest useful score.** MSP averages 70.0 / 62.9 / 65.0 AUROC (near / far / shift) over the four ID sets and is unstable across seeds. It falls below chance where the network is confidently wrong: far-OOD for Fashion-MNIST (48.3), MNIST digits for the SVHN model (31.9).
- **No per-image topological score wins on every ID set.** The local ΔH_0 score of [5] is best on average (84.6 / 89.2 / 89.1) but drops to 67.7 on KMNIST near-OOD, where simpler images (MNIST digits, Gaussian noise) lie *closer* to the training data than real KMNIST test images; the plain k-NN distance fails in the same way. The graph-based scores [2] and [4] are

ID set	near-OOD	far-OOD	covariate shift
MNIST	NotMNIST, F-MNIST, KMNIST	CIFAR10, SVHN, Textures, TIN, Places365, Gaussian, Uniform	MNIST-flip, MNIST-C
F-MNIST	MNIST, KMNIST, NotMNIST	CIFAR10, SVHN, Textures, TIN, Places365, Gaussian, Uniform	F-MNIST-flip
KMNIST	MNIST, F-MNIST, NotMNIST	CIFAR10, SVHN, Textures, TIN, Places365, Gaussian, Uniform	KMNIST-flip
SVHN	CIFAR10, TIN	Places365, Textures, F-MNIST, KMNIST, NotMNIST, Gaussian, Uniform	MNIST, SVHN-flip

Table 2: Part I: OOD sets for each in-distribution set (2000 test images each).

	MNIST (97.2%)			F-MNIST (87.2%)			KMNIST (85.3%)			SVHN (66.8%)			Mean over ID sets		
Method	Near	Far	Shift	Near	Far	Shift	Near	Far	Shift	Near	Far	Shift	Near	Far	Shift
MSP	74.1	69.2	75.3	72.2	48.3	66.3	73.4	72.6	71.9	60.2	61.4	46.4	70.0	62.9	65.0
Mutual information	80.9	78.6	79.3	80.9	57.7	81.0	**76.0**	77.3	77.5	66.7	61.1	50.8	76.1	68.7	72.2
k-NN distance†	89.6	90.6	78.1	91.5	87.7	95.8	65.2	73.6	75.6	75.4	**89.3**	83.6	80.4	85.3	83.3
[1] NP (per input)	59.2	53.5	57.0	72.4	63.8	68.7	55.8	51.8	56.4	51.0	53.8	51.4	59.6	55.7	58.4
[2] TU	80.7	82.6	65.3	81.6	72.9	66.9	71.2	80.4	61.4	50.9	67.1	67.6	71.1	75.8	65.3
[2] TU-rescaled	85.5	93.5	71.6	83.5	77.8	71.1	75.8	**90.5**	67.8	49.4	69.3	67.9	73.5	82.8	69.6
[3] pTU	87.0	84.3	75.0	47.2	63.6	75.0	55.9	68.6	75.0	63.9	59.4	59.2	63.5	69.0	71.0
[4] PL, full spectrum†	84.1	83.1	70.8	83.5	77.0	65.7	74.5	79.2	67.9	44.9	64.8	67.4	71.7	76.0	67.9
[4] PL, β_0 only†	86.1	84.6	75.5	77.7	77.1	64.5	73.5	79.7	74.3	44.4	62.4	65.6	70.4	75.9	70.0
[5] local $\Delta H_0{}^\dagger$	**96.1**	**97.0**	**84.3**	**96.2**	**92.4**	**98.7**	67.7	78.0	**89.3**	**78.2**	89.2	**84.1**	**84.6**	**89.2**	**89.1**

Table 3: Part I, per-image OOD detection: AUROC (%, mean over 3 seeds), averaged within each OOD group. ID test accuracy in parentheses. The first three rows are baselines; bold = best in the column. † = first 1000 images per set. Standard deviations and FPR@95 are in the repository.

strong for the three handwriting/clothing datasets but fall to chance for SVHN near-OOD, where natural photos produce activation graphs like those of house-number photos.

- **Activation-graph topology needs a well-trained model.** The graph-based scores degrade together with test accuracy (97% MNIST $\rightarrow$ 67% SVHN), consistent with the observation of [21] that a weak SVHN model "did not activate enough neurons".
- **pTU is not one-directional.** For Fashion-MNIST and KMNIST, OOD inputs such as MNIST digits and Gaussian noise have *lower* posterior variance than ID inputs (AUROC 17–25), so the one-sided score ranks them backwards. Its permutation test keeps the nominal false-alarm rate for MNIST (3–5%) but not for KMNIST (up to 28%).
- **Layer weighting matters.** Rescaling the layer distances of TU before averaging adds 2–11 AUROC points on every ID set.
- **Batches of inputs are easy for point-cloud topology.** With $m = 20$ images, [5] and [6] reach ≥ 97 AUROC on every ID set and OOD group except MNIST covariate shift (88–89) (Table 4).

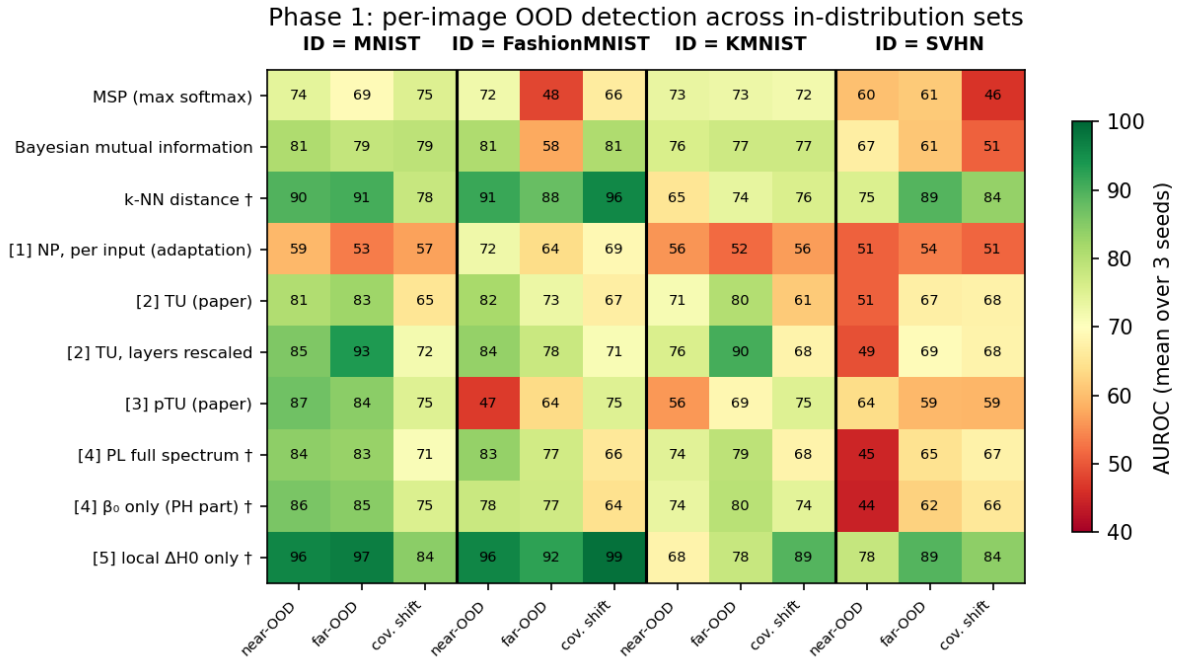


Figure 8: Per-image AUROC of every approach for each ID dataset and OOD group (mean over 3 seeds).

7 Part II: OpenOOD v1.5 CIFAR benchmarks

Setup. We use the OpenOOD v1.5 benchmarks and the released ResNet-18 checkpoints (three per benchmark): CIFAR-10 (near-OOD: CIFAR-100, TinyImageNet; far-OOD: MNIST, SVHN, Textures, Places365) and CIFAR-100 (near-OOD: CIFAR-10, TinyImageNet; same far-OOD), with OpenOOD's image lists, preprocessing and metrics. ResNet-18 has a single dense layer, so the graph-based methods [1]–[4] use the activation graph of fc : $\mathbb{R}^{512} \to \mathbb{R}^K$; the point-cloud methods [5]–[6] use the four residual stages and the pooled features. Methods marked † are evaluated on a fixed subsample of 2000 images per test set (1000 for [4]).

Reproducing the baselines. Our pipeline reproduces the published OpenOOD v1.5 numbers of MSP, EBO, MLS and KNN to two decimals, and ViM within one point (Table 5), so the topological methods are evaluated on exactly the published setting.

Findings.

- **On a well-trained ResNet-18, no per-image topological score beats the output-based baselines.** The best one, the β_0 curve of the fc activation graph ([4]), reaches 83.6 / 88.4 (near / far) on CIFAR-10 against 88.0 / 90.7 for MSP and 90.6 / 93.0 for KNN; on CIFAR-100 all graph-based scores stay around 67–71, about ten points below MSP. This reverses Part I, where the same scores beat MSP: the gains of topology appeared where the softmax of a small MLP was poorly calibrated, and they do not carry over to a strong convolutional network.
- **One dense layer is a narrow view.** ResNet-18 has a single dense layer, so [1]–[4] see only

m	Method	MNIST			F-MNIST			KMNIST			SVHN		
		N	F	S	N	F	S	N	F	S	N	F	S
10	MSP (batch mean)	93	79	**93**	91	33	89	90	86	91	79	70	44
	[2] TU (batch mean)	**98**	99	84	99	90	87	89	98	74	59	73	61
	[5] P_0, P_1, intrinsic dim.	87	**99**	83	99	99	**100**	98	100	**100**	91	**96**	**93**
	[6] SW to reference	83	98	82	**99**	**99**	100	**98**	**100**	100	**92**	95	88
	[6] layer profile	80	95	77	95	95	92	86	92	93	70	86	85
20	MSP (batch mean)	98	83	**98**	95	29	96	95	92	97	86	70	46
	[2] TU (batch mean)	**100**	99	90	100	94	94	93	100	80	63	74	56
	[5] P_0, P_1, intrinsic dim.	97	100	89	**100**	**100**	**100**	100	**100**	**100**	**100**	**99**	**100**
	[6] SW to reference	98	**100**	88	100	100	**100**	**100**	**100**	**100**	99	98	96
	[6] layer profile	92	99	79	98	98	98	93	97	92	84	94	88

Table 4: Part I, batch-level detection: AUROC (%) over batches of m images from one source (N = near-OOD, F = far-OOD, S = covariate shift; mean over 3 seeds).

the $512 \to K$ map from pooled features to logits, which carries the same information as the logits and the penultimate features that MSP, EBO and KNN already use. Within this view the ordering of Part I holds: β_0 and the full Laplacian spectrum (82–88 on CIFAR-10) are better than TU (74–80) and NP (72–75).

- **The pTU direction flips.** With a last-layer Laplace posterior, OOD inputs have *lower* topological variance than ID inputs (AUROC 25–35, i.e. 65–75 if the sign is reversed). The two-sided variant recovers 60–68 but remains weak, while the mutual information of the same posterior matches MSP (88.4 / 91.4 on CIFAR-10). pTU's behaviour depends strongly on the posterior approximation.

- **Local ΔH_0 on spatially averaged stage features is near chance (50–64).** In Part I it was the strongest per-image score, but there it acted on the raw input and small hidden layers. Here early convolutional stages, averaged over space, mostly encode low-level statistics; distances on normalized penultimate features (KNN, 90.6 / 93.0) are far more informative.

- **Batches:** on CIFAR-10 every batch score is near perfect (Table 6), including averaged MSP. On CIFAR-100, averaged MSP (96–99) and averaged TU (90–97) beat the point-cloud scores [5] and [6] (71–90): with 100 classes, a batch of 10–20 images covers the class structure too sparsely for its point-cloud topology to be stable.

8 Discussion

What topology adds, and when. In Part I, the inputs on which softmax confidence fails (natural images, textures and noise that a small MLP classifies confidently) are exactly where topological scores help most. In Part II, with a well-trained ResNet-18 whose only dense layer maps pooled features to logits, the topological scores see the same information as the output-based baselines and do not improve on them. Topology is therefore most useful when the network's own confidence is poorly calibrated, and when it can observe several dense layers; extending [1]–[4] to convolutional layers (e.g. through unrolled convolution graphs) is the main open step.

Simple versus rich summaries. The 0-dimensional information (MST weights, β_0 curves, local H_0 changes) carries most of the signal. Higher-dimensional features (H_1) and the nonzero persistent-

	CIFAR-10		CIFAR-100	
Method	Near	Far	Near	Far
Published (OpenOOD v1.5, Table 2)				
MSP	88.03	90.73	80.27	77.76
EBO	87.58	91.21	80.91	79.77
MLS	87.52	91.10	81.05	79.67
KNN	90.64	92.96	80.18	82.40
VIM	88.68	93.48	74.98	81.70
ReAct	87.11	90.42	80.77	80.39
ASH	75.27	78.49	78.20	80.58
RMDS	89.80	92.20	80.15	82.92
ODIN	82.87	87.96	79.90	79.28
MDS	84.20	89.72	58.69	69.39
This work (same checkpoints)				
MSP (ours)	88.03 ±0.25	90.73 ±0.43	80.27 ±0.11	77.76 ±0.44
EBO (ours)	87.58 ±0.46	91.21 ±0.92	80.91 ±0.08	79.77 ±0.61
MLS (ours)	87.52 ±0.47	91.10 ±0.89	81.05 ±0.07	79.67 ±0.57
KNN (ours)	90.64 ±0.20	92.96 ±0.14	80.18 ±0.15	82.40 ±0.17
ViM (ours)	88.73 ±0.43	93.27 ±0.20	75.92 ±0.16	82.14 ±0.65
Mutual information (Laplace)†	88.41 ±0.14	91.44 ±0.72	80.23 ±0.28	77.92 ±0.55
[1] NP (per input)	72.00 ±4.45	75.43 ±4.88	68.36 ±0.44	68.42 ±1.40
[2] TU	74.47 ±5.82	79.63 ±4.72	69.09 ±0.72	71.23 ±0.85
[3] pTU (Laplace)†	35.03 ±4.06	27.31 ±0.86	25.60 ±0.64	32.21 ±2.85
[3] pTU two-sided†	59.71 ±3.86	66.92 ±1.79	68.44 ±0.35	62.39 ±0.65
[4] PL, full spectrum†	82.36 ±1.14	87.36 ±1.33	70.77 ±0.60	69.69 ±0.31
[4] PL, β_0 only†	83.61 ±1.25	88.36 ±0.40	69.83 ±0.62	67.07 ±0.77
[5] local ΔH_0†	54.84 ±0.72	63.86 ±1.10	49.79 ±0.66	51.13 ±1.37

Table 5: Part II, OpenOOD v1.5 CIFAR benchmarks with the released ResNet-18 checkpoints: per-image AUROC (%, mean ± std over 3 checkpoints). The published rows are copied from OpenOOD v1.5; our re-implementations of the same baselines reproduce them. † = fixed subsample of 2000 images per test set (1000 for [4]).

Laplacian eigenvalues add little or hurt on these graphs, which are small, unweighted and triangle-free after thresholding.

Layers. The first layer of the MLP is the most informative for [2]–[4], while the published methods average layers equally; rescaling the layers improves TU by about 8 AUROC points in Part I.

Limitations. Part I uses one small MLP and grayscale data; Part II uses only the last dense layer of ResNet-18 for [1]–[4] and a last-layer Laplace posterior for [3]. Several scores are our adaptations (per-input NP, local ΔH_0, TU-rescaled, two-sided pTU, β_0 only) and are marked as such. The local ΔH_0 score is closely related to nearest-neighbour distances, so its strength should not be attributed to higher-order topology.

m	Method	CIFAR-10 Near	CIFAR-10 Far	CIFAR-100 Near	CIFAR-100 Far
10	[5] descriptors	99.0	97.9	71.6	79.9
	[6] SW to reference	98.8	98.8	70.8	84.5
	[6] layer profile	**99.6**	99.2	74.4	72.0
	MSP batch mean	98.9	**99.6**	**96.2**	**95.9**
	[2] TU batch mean	90.9	94.9	90.5	92.8
20	[5] descriptors	99.8	99.8	78.8	87.0
	[6] SW to reference	99.4	99.9	77.3	90.1
	[6] layer profile	**99.9**	100.0	80.3	81.8
	MSP batch mean	99.9	**100.0**	**97.9**	**99.1**
	[2] TU batch mean	92.6	96.4	94.7	96.9

Table 6: Part II, batch-level detection: AUROC (%) over batches of m images from one source (mean over 3 checkpoints).

9 Conclusion

We placed six topological approaches to analysing neural-network layers in one framework, implemented them on shared models and evaluated them as OOD detectors on four ID datasets and on the OpenOOD CIFAR benchmarks. On small MLPs, topology catches confident mistakes that output-based scores miss; on a well-trained ResNet-18, the topology of its single dense layer adds nothing over those scores. No per-image topological score wins everywhere; simple 0-dimensional summaries and, for few classes, batch-level point-cloud topology are the most robust.

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